

Maths - Question 1

Original paper page 1; original question and option regions

1. If $f : \mathbb{R} \rightarrow \mathbb{R}$, $g : \mathbb{R} \rightarrow \mathbb{R}$ are defined by $f(x) = 5x-3$, $g(x) = x^2 + 3$, then $(g \circ f^{-1})(3) =$

1) $\frac{25}{9}$

2) $\frac{111}{25}$

3) $\frac{9}{25}$

4) $\frac{25}{111}$

Sol: $f(x) = 5x-3$

$$f^{-1}(x) = \frac{x+3}{5}$$

$$f^{-1}(3) = \frac{6}{5}$$

$$g(f^{-1}(3)) = g\left(\frac{6}{5}\right) = \frac{36}{25} + 3 = \frac{36+75}{25}$$

$$= \frac{111}{25}$$

Maths - Question 2

Original paper page 1; original question and option regions

2. If $A = \left\{ x \in \mathbb{R} \mid \frac{\pi}{4} \leq x \leq \frac{\pi}{3} \right\}$ and $f(x) = \sin x - x$, then $f(A) =$

1) $\left[\frac{\sqrt{3}}{2} - \frac{\pi}{3}, \frac{1}{\sqrt{2}} - \frac{\pi}{4} \right]$

2) $\left[-\frac{1}{\sqrt{2}} - \frac{\pi}{4}, \frac{\sqrt{3}}{2} - \frac{\pi}{3} \right]$

3) $\left[-\frac{\pi}{3}, \frac{-\pi}{4} \right]$

4) $\left[\frac{\pi}{4}, \frac{\pi}{3} \right]$

Sol: $f(x) = \sin x - x$

$f'(x) = \cos x - 1 < 0$. f is $\downarrow$ (decreasing function)

Range is $\left[f\left(\frac{\pi}{3}\right), f\left(\frac{\pi}{4}\right) \right]$

$$\left[\frac{\sqrt{3}}{2} - \frac{\pi}{3}, \frac{1}{\sqrt{2}} - \frac{\pi}{4} \right]$$

Maths - Question 3

Original paper page 1; original question and option regions

3. The value of the sum $1.2.3 + 2.3.4 + 3.4.5 + \dots$ upto n terms =

1) $\frac{1}{6}n^2(2n^2 + 1)$

2) $\frac{1}{6}(n^2 - 1)(2n - 1)(2n + 3)$

3) $\frac{1}{8}(n^2 + 1)(n^2 + 5)$

4) $\frac{1}{4}n(n + 1)(n + 2)(n + 3)$

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Maths - Question 3 (continued)

Original paper page 2; original question and option regions

$$\begin{vmatrix} b^2 - ab & b - c & bc - ac \end{vmatrix}$$

Maths - Question 4

Original paper page 2; original question and option regions

4. The value of the determinant $\begin{vmatrix} ab - a^2 & a - b & b^2 - ab \\ bc - ac & c - a & ab - a^2 \end{vmatrix}$

1) abc

2) $a+b+c$

3) 0

4) $ab+bc+ca$

Sol: Give the values for a, b, c

Maths - Question 5

Original paper page 2; original question and option regions

5. If A is a square matrix of order 3, then $|\text{Adj}(\text{Adj}A^2)| =$

1) $|A|^2$

2) $|A|^4$

3) $|A|^8$

4) $|A|^{16}$

Sol: Use the formula. $|A^2|^4$

$$= |A|^8$$

Maths - Question 6

Original paper page 2; original question and option regions

6. The system $2x+3y+z=5$, $3x+y+5z=7$, $x+4y-2z=3$ has

1) Unique solution

2) Finite number of solutions

3) Infinite solutions

4) No solution

Sol: $2x+3y+z=5$ $2x+3y+z=5$ $3x+y+5z=7$ $9x+3y+15z=21$ $x+4y-2z=3$ $7x + 14z = 16 \Rightarrow x + 2z = 8$ $12x+4y+20z = 28$ $x + 4y - 2z = 3$ $11x+22z= 25$ $X+2z = 25/2$ $\therefore$ Parallel lines .

No solution.

Maths - Question 7

Original paper page 2; original question and option regions

$$7. \quad \sum_{k=1}^7 \left[\sin \frac{2k\pi}{7} - i \cos \frac{2k\pi}{7} \right] =$$

1) -1

2) 0

3) -i

4) i

$$\text{Sol:} \quad \sum_{k=1}^7 (-i) \left(\cos \frac{2k\pi}{7} + i \sin \frac{2k\pi}{7} \right)$$

$$(-i) \quad (-1) \quad = i$$

Maths - Question 8

Original paper page 2; original question and option regions

8. If 'ω' is a complex cube root of unity, then

$$\omega^{\left(\frac{1}{3} + \frac{2}{9} + \frac{4}{27} + \dots \infty\right)} + \omega^{\left(\frac{1}{2} + \frac{3}{8} + \frac{9}{32} + \dots \infty\right)} =$$

1) 1

2) -1

3) ω

4) i

Maths - Question 8 (continued)

Original paper page 3; original question and option regions

Sol: $\omega^{\frac{1}{3}\left(1+\frac{2}{(3)}+\frac{2^2}{(3)}+\dots\infty\right)} + \omega^{\frac{1}{2}\left(1+\frac{3}{4}+\frac{3^2}{(4)}+\dots\right)}$

$$\omega^{\frac{1}{3}\left[\frac{1}{1-\frac{2}{3}}\right]} + \omega^{\frac{1}{2}\left(\frac{1}{1-\frac{3}{4}}\right)}$$

$$\omega^{1/3\left(\frac{3}{1}\right)} + \omega^{\frac{1}{2}\left(\frac{4}{1}\right)}$$

$$\omega + \omega^2 = -1$$

Maths - Question 9

Original paper page 3; original question and option regions

9. The common roots of the equations $z^3 + 2z^2 + 2z + 1 = 0$, $z^{2014} + z^{1015} + 1 = 0$ are

1) ω, ω^2

2) $1, \omega, \omega^2$

3) $-1, \omega, \omega^2$

4) $-\omega, -\omega^2$

Sol: $z^3 + 2z^2 + 2z + 1 = 0$; $z^{2014} + z^{2015} + 1 = 0$

Verification.

$$\left(1 + \cos \frac{\pi}{8} - i \sin \frac{\pi}{8} \right)^8$$

Maths - Question 10

Original paper page 3; original question and option regions

10. $\left(\frac{1 + \cos \frac{\pi}{8} + i \sin \frac{\pi}{8}}{1 + \cos \frac{\pi}{8} + i \sin \frac{\pi}{8}} \right)^8 =$

1) 1

2) -1

3) 2

4) $\frac{1}{2}$

Sol: $\theta = \pi/8$ (say)

$$\left(\frac{1 + \cos \theta - i \sin \theta}{1 + \cos \theta + i \sin \theta} \right)^8$$

$$= \left(\frac{2 \cos^2 \frac{\theta}{2} - 2i \sin \frac{\theta}{2} \cos \frac{\theta}{2}}{2 \cos^2 \frac{\theta}{2} + 2i \sin \frac{\theta}{2} \cos \frac{\theta}{2}} \right)^8 = \left[\frac{\text{cis} \left(-\frac{\theta}{2} \right)}{\text{cis} (\theta/2)} \right]^8$$

$$= [\text{cis} (-\theta)]^8 = \cos 8\theta - i \sin 8\theta$$

$$\cos 8 \times \frac{\pi}{8} - i \sin 8 \times \frac{\pi}{8} = -1 - 0 = -1$$

Maths - Question 11

Original paper page 3; original question and option regions

11. If a, b, c are distinct and the roots of $(b-c)x^2 + (c-a)x + (a-b) = 0$ are equal, then a, b, c are in

1) Arithmetic progression

2) Geometric Progression

3) Harmonic progression

4) Arithmetico-Geometric progression

Sol: It is clear that, $x = 1$ is a root of the equation and $\therefore$ roots are equal

$$1 = \frac{a-b}{b-c}$$

$$b-c = a-b$$

$$2b = a+c \Rightarrow a, b, c \text{ are in AP}$$

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Maths - Question 11 (continued)

Original paper page 4; original question and option regions

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Maths - Question 12

Original paper page 4; original question and option regions

12. If the roots of $x^3 - kx^2 + 14x - 8 = 0$ are in geometric progression, then $k =$

1) -3

2) 7

3) 4

4) 0

Sol: $x^3 - kx^2 + 14x - 8 = 0$

$\frac{a}{r}, a, ar$

Product of roots = +8

$$a^3 = +8$$

$$a = +2$$

sub in the above

$$+8 - 4k + 28 - 8 = 0$$

$$4k = 28$$

$$k = 7$$

Maths - Question 13

Original paper page 4; original question and option regions

13. If the harmonic mean of the roots of $\sqrt{2}x^2 - bx + (8 - 2\sqrt{5}) = 0$ is 4, then the value of b=

1) 2

2) 3

3) $4 - \sqrt{5}$

4) $4 + \sqrt{5}$

Sol: $\alpha + \beta = \frac{b}{\sqrt{2}}; \alpha\beta = \frac{8 - 2\sqrt{5}}{\sqrt{2}}$

$$\frac{2\alpha\beta}{\alpha + \beta} = 4$$

$$\frac{\sqrt{2}(8 - 2\sqrt{5})\sqrt{2}}{b} = 4$$

$$2b = 8 - 2\sqrt{5} \Rightarrow b = 4 - \sqrt{5}$$

Maths - Question 14

Original paper page 4; original question and option regions

14. For real values of x, the range of $\frac{x^2 + 2x + 1}{x^2 + 2x - 1}$ is

- 1) $(-\infty, 0) \cup (1, \infty)$ 2) $\left[\frac{1}{2}, 2\right]$ 3) $\left(-\infty, \frac{-2}{9}\right] \cup (1, \infty)$ 4) $(-\infty, -6] \cup (-2, \infty)$

Sol: $\frac{x^2 + 2x + 1}{x^2 + 2x - 1} = y$

Simplify, and $\text{disc} \geq 0$

Maths - Question 15

Original paper page 4; original question and option regions

15. The number of four digit numbers formed by using the digits 0,2,4,5 and which are not divisible by 5, is

1) 10

2) 8

3) 6

4) 4

Sol: $_{- - -} \underline{0} = 3! = 6$

$_{- - -} \underline{5} = 2 \times 2 \times 1 = 4$

Total 10 ways

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Maths - Question 15 (continued)

Original paper page 5; original question and option regions

|

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Maths - Question 16

Original paper page 5; original question and option regions

16. T_m denotes the number of Triangles that can be formed with the vertices of a regular polygon of m sides. If $T_{m+1} - T_m = 15$, then $m =$

1) 3

2) 6

3) 9

4) 12

Sol: $m+1C_3 - mC_3 = 15$

Verification $m = 6$; $7C_3 - 6C_3$

$$35 - 20 = 15$$

Maths - Question 17

Original paper page 5; original question and option regions

17. If $|x| < 1$ then the coefficient of x^5 in the expansion of $\frac{3x}{(x-2)(x+1)}$ is

1) $\frac{33}{32}$

2) $-\frac{33}{32}$

3) $\frac{31}{32}$

4) $-\frac{31}{32}$

Sol: $\frac{3x}{(x-2)(x+1)} = \frac{A}{x-2} + \frac{B}{x+1} =$

$$x=2; \quad A = \frac{6}{3} = 2$$

$$x = -1 \quad B = \frac{-3}{-3} = 1$$

$$\frac{2}{(-2)\left[1 - \frac{x}{2}\right]} + \frac{1}{1+x}$$

$$-1\left(1 - \frac{x}{2}\right)^{-1} + (1+x)^{-1}$$

$$\left[1 + \frac{x}{2} + \frac{x^2}{4} + \frac{x^3}{8} + \frac{x^4}{16} + \frac{x^5}{32}\right] + \left[1 - x + x^2 - x^3 + x^4 - x^5\right]$$

$$-\frac{1}{32} - 1 = -33/32$$

Maths - Question 18

Original paper page 5; original question and option regions

18. If the coefficients of x^9, x^{10}, x^{11} in the expansion of $(1+x)^n$ are in arithmetic progression then

$$n^2 - 41n =$$

1) 398

2) 298

3) -398

4) 198

Sol: $(n-2r)^2 = n+2$

$${}^nC_9, {}^nC_{10}, {}^nC_{11} \text{ are in AP}$$

$$r = 10$$

$$(n-20)^2 = n+2$$

$$n^2 + 400 - 40n = n + 2$$

$$n^2 - 41n = -398$$

Maths - Question 19

Original paper page 5; original question and option regions

19. If $x = \frac{1}{5} + \frac{1.5}{5.10} + \frac{1.5.5}{5.10.15} + \dots \infty$, then $3x^2 + 6x =$

1) 1

2) 2

3) 3

4) 4

Maths - Question 19 (continued)

Original paper page 6; original question and option regions

$$\text{Sol: } 1+x = 1 + \frac{1}{1!} \frac{1}{5} + \frac{1.3}{2!} \left(\frac{1}{5}\right)^2 + \dots$$

$$S = (1-x)^{-p/q}; P=1, q=2$$

$$\frac{x}{q} = \frac{1}{5}$$

$$x = \frac{2}{5}$$

$$(1+x) = \left(1 - \frac{2}{5}\right)^{-1/2}$$

$$(1+x) = \left(\frac{3}{5}\right)^{-1/2} = \left(\frac{5}{3}\right)^{1/2}$$

$$1+x^2+2x = \frac{5}{3}$$

$$3x^2 + 6x + 3 = 5$$

$$3x^2 + 6x = 2$$

Maths - Question 20

Original paper page 6; original question and option regions

20. If $\sin \theta + \cos \theta = p$ and $\tan \theta + \cot \theta = q$ then $q(p^2 - 1) =$

1) $\frac{1}{2}$

2) 2

3) 1

4) 3

Sol: $\sin \theta + \cos \theta = P, \frac{1}{\sin \theta \cos \theta} = q$

$$\sin \theta \cos \theta = 1/q$$

$$1 + \frac{2}{q} = P^2 \Rightarrow \frac{2}{q} = p^2 - 1$$

$$\Rightarrow 2 = q(p^2 - 1) \therefore q(p^2 - 1) = 2$$

Maths - Question 21

Original paper page 6; original question and option regions

21. $\tan \frac{\pi}{5} + 2 \tan \frac{2\pi}{5} + 4 \cot \frac{4\pi}{5} =$

1) $\cot \frac{\pi}{5}$

2) $\cot \frac{2\pi}{5}$

3) $\cot \frac{3\pi}{5}$

4) $\cot \frac{4\pi}{5}$

Sol: $\tan \theta + 2 \tan 2\theta + 4 \tan 4\theta = \cot \theta$

$$\tan \frac{\pi}{5} + 2 \tan \frac{2\pi}{5} + 4 \tan \frac{4\pi}{5} = \cot \frac{\pi}{5}$$

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Maths - Question 21 (continued)

Original paper page 7; original question and option regions

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Maths - Question 22

Original paper page 7; original question and option regions

22. If $\sin A + \sin B + \sin C = 0$ and $\cos A + \cos B + \cos C = 0$,

then $\cos(A+B) + \cos(B+C) + \cos(C+A) =$

1) $\cos(A+B+C)$

2) 2

3) 1

4) 0

Sol: Use complex numbers

$$x = \cos A + i \sin A$$

$$y = \cos B + i \sin B$$

$$z = \cos C + i \sin C$$

$$\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 0$$

$$yz + zx + xy = 0$$

$$\cos(B+C) + \cos(C+A) + \cos(A+B) = 0$$

$$\Rightarrow \cos(B+C) + \cos(C+A) + \cos(A+B) = 0$$

Maths - Question 23

Original paper page 7; original question and option regions

23. If $\tan \theta \cdot \tan(120^\circ - \theta) \cdot \tan(120^\circ + \theta) = \frac{1}{\sqrt{3}}$, then $\theta =$

- 1) $\frac{n\pi}{3} + \frac{\pi}{18}, n \in \mathbb{Z}$ 2) $\frac{n\pi}{3} + \frac{\pi}{12}, n \in \mathbb{Z}$ 3) $\frac{n\pi}{12} + \frac{\pi}{12}, n \in \mathbb{Z}$ 4) $\frac{n\pi}{3} + \frac{\pi}{6}, n \in \mathbb{Z}$

Sol: $\tan 3\theta = \tan \frac{\pi}{6} \Rightarrow 3\theta = n\pi + \frac{\pi}{6}$

/ ...2 ...3 \ / ...4 ...6 \ _

Maths - Question 24

Original paper page 7; original question and option regions

24. If $\sin^{-1}\left(x - \frac{x}{2} + \frac{x}{4} - \dots \infty\right) + \cos^{-1}\left(x^2 - \frac{x}{2} + \frac{x}{4} - \dots \infty\right) = \frac{\pi}{2}$ **and** $0 < x < \sqrt{2}$ **then x =**

1) $\frac{1}{2}$

2) 1

3) $-\frac{1}{2}$

4) -1

Sol: $x = x^2 \Rightarrow x[x-1] = 0 \Rightarrow x = 1$

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Maths - Question 25

Original paper page 7; original question and option regions

25. If $2 \sinh^{-1} \left(\frac{a}{\sqrt{1-a^2}} \right) = \log \left(\frac{1+x}{1-x} \right)$ then $x =$

- 1) a 2) $\frac{1}{a}$ 3) $\sqrt{1-a^2}$ 4) $\frac{1}{\sqrt{1-a^2}}$

Sol:
$$2 \cos \left[\frac{a}{\sqrt{1-a^2}} + \sqrt{1 + \frac{a^2}{1-a^2}} \right] = 2 \cos \left[\frac{a+1}{\sqrt{1-a^2}} \right] =$$

$$= 2 \cos \sqrt{\frac{1+a}{1-a}} = \cos \left[\frac{1-a}{1-a} \right]$$

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Maths - Question 25 (continued)

Original paper page 8; original question and option regions

|

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Maths - Question 26

Original paper page 8; original question and option regions

26. In a $\triangle ABC$, $(a+b+c)(b+c-a) = \lambda bc$, then

1) $\lambda < -6$

2) $\lambda > 6$

3) $0 < \lambda < 4$

4) $\lambda > 4$

Sol: $2s \cdot 2(s-a) = \lambda bc \Rightarrow \frac{s(s-a)}{bc} = \frac{\lambda}{4}$

$$\Rightarrow \cos^2 \frac{A}{2} = \frac{\lambda}{4} \quad \Rightarrow 0 \leq \frac{\lambda}{4} \leq 1$$

$$\Rightarrow 0 < \lambda < 4$$

Maths - Question 27

Original paper page 8; original question and option regions

27. If in a $\triangle ABC$, $r_1 = 2r_2 = 3r_3$, then the perimeter of the triangle is equal to

1) $3a$

2) $3b$

3) $3c$

4) $3(a+b+c)$

Sol: $a : b : c = (2+3) : (1+3) : (2+1) = 5 : 4 : 3$

$$2S = 12K = 3(4K) \quad (\text{or}) \quad 4(3K)$$

$$= 3b \quad (\text{or}) \quad 4C$$

Maths - Question 28

Original paper page 8; original question and option regions

28. In ΔABC , $\frac{a}{\tan A} + \frac{b}{\tan B} + \frac{c}{\tan C} =$

1) $2r$

2) $r+2R$

3) $2r+R$

4) $2(r+R)$

Sol: $2R (\cos A + \cos B + \cos C) = 2R \left[1 + \frac{r}{R} \right] = 2R + 2r$

Maths - Question 29

Original paper page 8; original question and option regions

29. If m_1, m_2, m_3, m_4 are respectively the magnitudes of the vectors

$\vec{a}_1 = 2\vec{i} - \vec{j} + \vec{k}, \vec{a}_2 = 3\vec{i} - 4\vec{j} - 4\vec{k}, \vec{a}_3 = -\vec{i} + \vec{j} - \vec{k}, \vec{a}_4 = -\vec{i} + 3\vec{j} + \vec{k}$, then the correct order of

m_1, m_2, m_3, m_4 is

- 1) $m_3 < m_1 < m_4 < m_2$ 2) $m_3 < m_1 < m_2 < m_4$ 3) $m_3 < m_4 < m_1 < m_2$ 4) $m_3 < m_4 < m_2 < m_1$

Sol: $|\vec{a}_1| = \sqrt{6}, |\vec{a}_2| = \sqrt{41}, |\vec{a}_3| = \sqrt{3}, |\vec{a}_4| = \sqrt{11}$

$$m_3 < m_1 < m_4 < m_2$$

Maths - Question 30

Original paper page 8; original question and option regions

30. If $\bar{a}, \bar{b}, \bar{c}$ are unit vectors such that $\bar{a} + \bar{b} + \bar{c} = \bar{0}$, then the $\bar{a} \cdot \bar{b} + \bar{b} \cdot \bar{c} + \bar{c} \cdot \bar{a} =$

1) $\frac{3}{2}$

2) $-\frac{3}{2}$

3) $\frac{1}{2}$

4) $-\frac{1}{2}$

Sol: $1+1+1+2 \quad \sum \bar{a} \cdot \bar{b} = 0 \Rightarrow \sum \bar{a} \cdot \bar{b} = -3/2$

Maths - Question 31

Original paper page 8; original question and option regions

31. If $\vec{a} = 2\vec{i} + \vec{k}, \vec{b} = \vec{i} + \vec{j} + \vec{k}, \vec{c} = 4\vec{i} - 3\vec{j} + 7\vec{k}$ then the vector $\vec{r}$ satisfying $\vec{r} \times \vec{b} = \vec{c} \times \vec{b}$ and $\vec{r} \cdot \vec{a} = 0$ is

1) $\vec{i} + 8\vec{j} + 2\vec{k}$

2) $\vec{i} - 8\vec{j} + 2\vec{k}$

3) $\vec{i} - 8\vec{j} - 2\vec{k}$

4) $-\vec{i} - 8\vec{j} + 2\vec{k}$

Sol: $(\vec{r} - \vec{c}) \times \vec{b} = \vec{0} \Rightarrow \vec{r} = \vec{c} + t\vec{b}$

Maths - Question 31 (continued)

Original paper page 9; original question and option regions

$$\Rightarrow \vec{r} \cdot \vec{a} = 0 \Rightarrow \vec{c} \cdot \vec{a} + t(\vec{b} \cdot \vec{a}) = 0 \Rightarrow t = -\frac{\vec{c} \cdot \vec{a}}{\vec{b} \cdot \vec{a}}$$

$$\vec{r} = \vec{c} - \frac{\vec{c} \cdot \vec{a}}{\vec{b} \cdot \vec{a}}(\vec{b}) = (4, -3, 7) - \left(\frac{15}{3}\right)(1, 1, 1) \Rightarrow (-1, -8, 2)$$

Maths - Question 32

Original paper page 9; original question and option regions

32. If $\vec{a}, \vec{b}, \vec{c}$ are three vectors such that $|\vec{a}|=1, |\vec{b}|=2, |\vec{c}|=3$, and $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c} = \vec{c} \cdot \vec{a} = 0$, then $\left| \begin{bmatrix} \vec{a} & \vec{b} & \vec{c} \end{bmatrix} \right| =$

1) 0

2) 2

3) 3

4) 6

Sol:
$$\left(\begin{bmatrix} \vec{a} & \vec{b} & \vec{c} \end{bmatrix} \right)^2 = \begin{vmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 9 \end{vmatrix} = 36 \Rightarrow \left(\begin{bmatrix} \vec{a} & \vec{b} & \vec{c} \end{bmatrix} \right) = 6$$

Maths - Question 33

Original paper page 9; original question and option regions

33. If $\left[\bar{a} \times \bar{b} \quad \bar{b} \times \bar{c} \quad \bar{c} \times \bar{a}\right] = \lambda \left[\bar{a} \quad \bar{b} \quad \bar{c}\right]^2$, then $\lambda =$

1) 0

2) 1

3) 2

4) 3

Sol: $\left[\bar{a} \times \bar{b} \quad \bar{b} \times \bar{c} \quad \bar{c} \times \bar{a}\right] = \left[\bar{a} \quad \bar{b} \quad \bar{c}\right]^2$

$$\lambda = 1$$

Maths - Question 34

Original paper page 9; original question and option regions

34. The Cartesian equation of the plane passing through the point (3, -2, -1) and parallel to the vectors $\vec{b} = \vec{i} - 2\vec{j} + 4\vec{k}$ and $\vec{c} = 3\vec{i} + 2\vec{j} - 5\vec{k}$

- 1) $2x - 17y - 8z + 63 = 0$ 2) $3x + 17y + 8z - 36 = 0$ 3) $2x + 17y + 8z + 36 = 0$ 4) $3x - 16y + 8z - 63 = 0$

Sol: Sol:
$$\begin{vmatrix} x-3 & y+2 & z+1 \\ 1 & -2 & 4 \\ 3 & 2 & -5 \end{vmatrix} = 0$$

$$\Rightarrow (2x - 6) + 17y + 34 + 8z + 8 = 0$$

$$\Rightarrow 2x + 17y + 8z + 36 = 0$$

Ans (3)

Maths - Question 35

Original paper page 9; original question and option regions

35. The arithmetic mean of the observations 10, 8, 5, a, b is 6 and their variance is 6.8. Then ab=

1) 6

2) 4

3) 3

4) 12

Sol: $AM = \frac{23 + a + b}{5} = 6 \Rightarrow a + b = 7$

$$\text{Variance} = \frac{16 + 4 + 1 + (a - 6)^2 + (b - 6)^2}{5} = 6.8$$

$$\therefore a^2 + b^2 = 25 \Rightarrow ab = 12$$

Ans (4)

Maths - Question 36

Original paper page 9; original question and option regions

36. If the median of the data 6, 7, x-2, x, 18, 21 written in ascending order is 16, then the variance of that data is

1) $30\frac{1}{5}$

2) $31\frac{1}{3}$

3) $32\frac{1}{2}$

4) $33\frac{1}{3}$

Maths - Question 36 (continued)

Original paper page 10; original question and option regions

$$\text{Sol: Median} = 16 \Rightarrow x = 17$$

$$\text{Variance} = \frac{64 + 49 + 1 + 9 + 16 + 49}{6} = 31 \frac{1}{3}$$

Ans (2)

Maths - Question 37

Original paper page 10; original question and option regions

37. Two persons A and B are throwing an unbiased six faced die alternatively, with the condition that the person who throws 3 first wins the game. If A starts the game, the probabilities of A and B to win the same are respectively

1) $\frac{6}{11}, \frac{5}{11}$

2) $\frac{5}{11}, \frac{6}{11}$

3) $\frac{8}{11}, \frac{3}{11}$

4) $\frac{3}{11}, \frac{8}{11}$

Sol: $p = \frac{1}{6}, q = \frac{5}{6}$

$$P(\text{A's win}) = p + q^2p + q^4p + \dots = p \left(\frac{1}{1 - q^2} \right) = \frac{1}{1 + q} = \frac{6}{11}$$

Ans $\frac{6}{11}, \frac{5}{11}$

Ans (1)

Maths - Question 38

Original paper page 10; original question and option regions

38. The letters of the word “QUESTION” are arranged in a row at random. The probability that there are exactly two letters between Q and S is

1) $\frac{1}{14}$

2) $\frac{5}{7}$

3) $\frac{1}{7}$

4) $\frac{5}{28}$

Sol:

1 2 3 4 5 6 7 8

QUESTION

(1,4) (2,5) (3,6) (4,7) (5,8) $\rightarrow$ 5 positions $\rightarrow$ 2 ways

Remaining letters be arranged in 6 ! ways

$$P = \frac{5 \times 2 \times 6!}{8!} = \frac{5}{28}$$

Ans (4)

Maths - Question 39

Original paper page 10; original question and option regions

39. If $\frac{1+3p}{3}, \frac{1-2p}{2}$ are probabilities of two mutually exclusive events, then p lies in the interval

1) $\left[-\frac{1}{3}, \frac{1}{2}\right]$

2) $\left(-\frac{1}{2}, \frac{1}{2}\right)$

3) $\left[-\frac{1}{3}, \frac{2}{3}\right]$

4) $\left(-\frac{1}{3}, \frac{2}{3}\right)$

Sol.

$$\begin{array}{l}
 0 \leq \frac{1+3p}{3} \leq 1 \quad \left| \quad 0 \leq \frac{1-2p}{2} \leq 1 \right| \quad 0 \leq \frac{1+3p}{3} + \frac{1-2p}{2} \leq 1 \\
 0 \leq 1+3p \leq 3 \quad \left| \quad 0 \leq 1-2p \leq 2 \right| \\
 -1 \leq 3p \leq 2 \quad \left| \quad -1 \leq -2p \leq 1 \right| \quad 0 < \frac{5}{6} < 1 \\
 -\frac{1}{3} \leq p \leq \frac{2}{3} \quad \left| \quad -\frac{1}{2} \leq p \leq \frac{1}{2} \right| \\
 \qquad \qquad \qquad -\frac{1}{3} \leq p \leq \frac{1}{2}
 \end{array}$$

Ans (1)

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Maths - Question 39 (continued)

Original paper page 11; original question and option regions

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Maths - Question 40

Original paper page 11; original question and option regions

40. The probability that an event does not happen in one trial is 0.8. The probability that the event happens atmost once in three trials is

1) 0.896

2) 0.791

3) 0.642

4) 0.592

Sol. $q = \frac{8}{10} = \frac{4}{5}, p = \frac{1}{5}$

$$P(X \leq 1) = P(X = 0) + P(X = 1)$$

Ans (1)

$$= {}^3C_0(1)\left(\frac{4}{5}\right)^3 + {}^3C_1\frac{1}{5}\left(\frac{4}{5}\right)^2$$

$$= 0.896$$

Maths - Question 41

Original paper page 11; original question and option regions

41. The probability distribution of a random variable is given below:

$X=x$	0	1	2	3	4	5	6	7
$P(X=x)$	0	K	$2K$	$2K$	$3K$	K^2	$2K^2$	$7K^2 + K$

Then $P(0 < X < 5)$

1) $\frac{1}{10}$

2) $\frac{3}{10}$

3) $\frac{8}{10}$

4) $\frac{7}{10}$

Sol: $10K^2 + 9K = 1 \Rightarrow K(10K + 9) = 1$

$$\Rightarrow 10K^2 + 9K - 1 = 0 \Rightarrow K = \frac{1}{10}$$

$$P(0 < x < 5) = P(x=1) + P(x=2) + P(x=3) + P(x=4)$$

$$= 8K = \frac{8}{10} = 0.8$$

Maths - Question 42

Original paper page 11; original question and option regions

42. If the equation to the locus of points equidistant from the points $(-2,3)$, $(6,-5)$ is $ax+by+c=0$ where $a > 0$ then, the ascending order of a, b, c is

1) a, b, c 2) c, b, a 3) b, c, a 4) a, c, b

Sol: $A = (-2, 3)$ $B = (6, -5)$

Mid point of $AB = (2, -1)$

$$\text{Slope of } AB = \frac{-8}{8} = -1$$

Equ of perpendicular bisector of AB in

$$y+1 = 1(x-2) \Rightarrow x - y - 3 = 0$$

Comparing with $ax+by+c=0$

$$\frac{a}{1} = \frac{b}{-1} = \frac{c}{-3} = k \Rightarrow a = k, b = -k, c = -3k$$

$$\Rightarrow c < b < a$$

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Maths - Question 42 (continued)

Original paper page 12; original question and option regions

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Maths - Question 43

Original paper page 12; original question and option regions

43. The point (2, 3) is first reflected in the straight line $y = x$ and then translated through a distance of 2 units along the positive direction of x-axis. The coordinates of the transformed point are

1) (5, 4)

2) (2, 3)

3) (5, 2)

4) (4, 5)

Sol: If (2, 3) in the line $x = y$ in (3, 2) after shifting 2 units along positive x-axis it becomes (5, 2)

Maths - Question 44

Original paper page 12; original question and option regions

44. If the straight lines $2x+3y-1=0$, $x+2y-1=0$ and $ax+by-1=0$ form a triangle with origin as ortho centre, then (a, b) =

1) $(6, 4)$

2) $(-3, 3)$

3) $(-8, 8)$

4) $(0, 7)$

Sol: $(a_1a_2 + b_1b_2) L_3 = (a_1a_3 + b_1b_3) L_2 = (a_2a_3 + b_2b_3) L_1$

$$8(ax+by-1) = (2a+3b)(x+2y-1) = (a+2b)(2x+3y-1) \text{ it satisfies by orthocenter}$$

$$\text{Sub } (0, 0), -8 = -2a - 3b = -a - 2b$$

$$2a + 3b = 8$$

$$a + 2b = 8$$

$$\text{Solving } a = -8, b = 8$$

Maths - Question 45

Original paper page 12; original question and option regions

45. The point on the line $4x - y - 2 = 0$ which is equidistant from the points $(-5, 6)$ and $(3, 2)$ is

1) $(2, 6)$

2) $(4, 14)$

3) $(1, 2)$

4) $(3, 10)$

Sol: Equation of perpendicular bisector of line joining $(-5, 6)$, $(3, 2)$ is

$$= 2x(-8) + 2y(4) = 25 + 36 - 9 - 4 \Rightarrow -16x + 8y = 48$$

$$\Rightarrow 2x - y + 6 = 0 \quad \text{Solving with } 4x - y - 2 = 0$$

$$2x - 8 = 0$$

$$x = 4, y = 14$$

Maths - Question 46

Original paper page 12; original question and option regions

46. If the lines $x + 2ay + a = 0$, $x + 3by + b = 0$, $x + 4cy + c = 0$ are concurrent, then a, b, c are in

1) Arithmetic progression

2) Geometric progression

3) Harmonic progression

4) Arithmetico – Geometric progression

Sol:
$$\begin{vmatrix} 1 & 2a & a \\ 1 & 3b & b \\ 1 & 4c & c \end{vmatrix} = 0$$

$$\Rightarrow 1(3bc - 4bc) - (2ac - 4ac) + 1(2ab - 3ab) = 0$$

$$\Rightarrow -bc + 2ac - ab = 0 \Rightarrow ab + bc = 2ac$$

$$\Rightarrow \frac{1}{c} + \frac{1}{a} = \frac{2}{b}$$

$\therefore a, b, c$ are in H.P.

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Maths - Question 46 (continued)

Original paper page 13; original question and option regions

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Maths - Question 47

Original paper page 13; original question and option regions

47. The angle between the straight lines represented by $(x^2 + y^2)\sin^2 \alpha = (x \cos \alpha - y \sin \alpha)^2$ is

1) $\frac{\alpha}{2}$

2) α

3) 2α

4) $\frac{\pi}{2}$

Sol: Given equation $x^2 \cos 2\alpha - xy \sin 2\alpha = 0$

$$\cos \theta = \frac{(\cos 2\alpha + 0)}{\sqrt{\cos^2 2\alpha + \sin^2 2\alpha}} \Rightarrow \theta = 2\alpha$$

Maths - Question 48

Original paper page 13; original question and option regions

48. If the slope of one of the lines represented by $ax^2 - 6xy + y^2 = 0$ is the square of the other, then the value of a is

1) -27 or 8

2) -3 or 2

3) -64 or 27

4) -4 or 3

Sol: $m + m^2 = \frac{6}{1}, m \cdot m^2 = \frac{a}{1}$

$$m^3 + m^6 + 3mm^2(m + m^2) = 216$$

$$\Rightarrow a + a^2 + 3a(6) = 216 \Rightarrow a^2 + 19a - 216 = 0$$

$$\Rightarrow a = -27 \text{ or } 8$$

Maths - Question 49

Original paper page 13; original question and option regions

49. The sum of the minimum and maximum distances of the point (4, -3) to the circle

$$x^2 + y^2 + 4x - 10y - 7 = 0 \text{ is}$$

1) 10

2) 12

3) 16

4) 20

Sol: $P = (4, -3)$, centre of circle $C = (-2, 5)$

Radius $r = 6$

Minimum distance of $P = CP - r$

Maximum distance of $P = CP + r$

$$\text{Sum} = 2CP = 2\sqrt{36 + 64} = 20$$

Maths - Question 50

Original paper page 13; original question and option regions

50. The locus of centres of the circles which cut the circles $x^2 + y^2 + 4x - 6y + 9 = 0$

$x^2 + y^2 - 5x + 4y + 2 = 0$ orthogonally is

- 1) $3x + 4y - 5 = 0$ 2) $9x - 10y + 7 = 0$ 3) $9x + 10y - 7 = 0$ 4) $9x - 10y + 11 = 0$

Sol: Required locus is radical axis of given circles

i.e $9x - 10y + 7 = 0$

Maths - Question 51

Original paper page 13; original question and option regions

51. The equation of the circle passing through (2, 0) and (0, 4) and having the minimum radius is

1) $x^2 + y^2 = 20$

2) $x^2 + y^2 - 2x - 4y = 0$

3) $x^2 + y^2 = 4$

4) $x^2 + y^2 = 16$

Maths - Question 51 (continued)

Original paper page 14; original question and option regions

Sol: Required circle is a circle on (2,0), (0,4) as ends of diameter

$$\text{i.e } (x-2)(x-0) + (y-0)(y-4) = 0$$

$$\Rightarrow x^2 + y^2 - 2x - 4y = 0$$

Maths - Question 52

Original paper page 14; original question and option regions

52. If $x^2 + y^2 - 4x - 2y + 5 = 0$ and $x^2 + y^2 - 6x - 4y - 3 = 0$ are members of a coaxal system of circles then centre of a point circle in the system is

1) $(-5, -6)$

2) $(5, 6)$

3) $(3, 5)$

4) $(-8, -13)$

Sol: Equation of coaxal system of circles is $S + \lambda L = 0$

$$\Rightarrow x^2 + y^2 - 4x - 2y + 5 + 2\lambda(x + y + 4) = 0$$

$$C = (-\lambda + 2, -\lambda + 1), r = \sqrt{(-\lambda + 2)^2 + (-\lambda + 1)^2 - 5 - 8\lambda}$$

$$r = 0 \Rightarrow (\lambda - 2)^2 + (\lambda - 1)^2 - 5 - 8\lambda = 0 \Rightarrow 2\lambda^2 - 14\lambda = 0 \Rightarrow \lambda = 0, 7$$

$$C = (-5, -6)$$

Maths - Question 53

Original paper page 14; original question and option regions

53. If $x - y + 1 = 0$ meets the circle $x^2 + y^2 + y - 1 = 0$ at A and B then the equation of the circle with AB as diameter is

1) $2(x^2 + y^2) + 3x - y + 1 = 0$

2) $2(x^2 + y^2) + 3x - y + 2 = 0$

3) $2(x^2 + y^2) + 3x - y + 3 = 0$

4) $x^2 + y^2 + 3x - y + 1 = 0$

Sol: Concept

Maths - Question 54

Original paper page 14; original question and option regions

54. An equilateral triangle is inscribed in the parabola $y^2 = 8x$, with one of its vertices is the vertex of the parabola. Then, the length of the side of that triangle is

1) $24\sqrt{3}$

2) $16\sqrt{3}$

3) $8\sqrt{3}$

4) $4\sqrt{3}$

Sol: $\tan 30^\circ = \frac{2at}{at^2} \Rightarrow \frac{1}{\sqrt{3}} = \frac{2}{t} \Rightarrow t = 2\sqrt{3}$

Length of side = $4at = 4(2)(2\sqrt{3}) = 16\sqrt{3}$

Maths - Question 55

Original paper page 14; original question and option regions

55. The point (3, 4) is the focus and $2x - 3y + 5 = 0$ is the directrix of a parabola. Its latus rectum is

1) $\frac{2}{\sqrt{13}}$

2) $\frac{4}{\sqrt{13}}$

3) $\frac{1}{\sqrt{13}}$

4) $\frac{3}{\sqrt{13}}$

Sol: Distance from focus (3,4) to directrix $2x - 3y + 5 = 0$

$$= \frac{|6 - 12 + 5|}{\sqrt{13}} = \frac{1}{\sqrt{13}}$$

$$\therefore 2a = \frac{1}{\sqrt{13}}$$

$$\text{Length of latus rectum is } = 4a = \frac{2}{\sqrt{13}}$$

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Maths - Question 55 (continued)

Original paper page 15; original question and option regions

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$\sqrt{2}$

$\sqrt{2}$

|

Maths - Question 56

Original paper page 15; original question and option regions

56. The radius of the circle passing through the foci of the ellipse $\frac{x}{16} + \frac{y}{9} = 1$ and having its centre at (0,3) is

1) 6

2) 4

3) 3

4) 2

Sol: Centre of circle = (0,3) is B, the end of minor axis

The circle passes through the focus S

$\therefore$ radius of circle = SB = a = 4

Maths - Question 57

Original paper page 15; original question and option regions

57. The values that m can take so that the straight line $y = 4x + m$ touches the curve $x^2 + 4y^2 = 4$ is

1) $\pm\sqrt{45}$

2) $\pm\sqrt{60}$

3) $\pm\sqrt{65}$

4) $\pm\sqrt{72}$

Sol: Ellipse is $\frac{x^2}{4} + \frac{y^2}{1} = 1$

It touches $y = 4x + m$

$$\therefore c^2 = a^2m^2 + b^2 \Rightarrow m^2 = 4(4)^2 + 1 = 65$$

$$m = \pm\sqrt{65}$$

Maths - Question 58

Original paper page 15; original question and option regions

58. The foci of the ellipse $\frac{x^2}{16} + \frac{y^2}{b^2} = 1$ and the hyperbola $\frac{x^2}{144} - \frac{y^2}{81} = \frac{1}{25}$ coincide. Then , the value of b^2 is

1) 5

2) 7

3) 9

4) 1

Sol: $\frac{x^2}{16} + \frac{y^2}{b^2} = 1, \quad \frac{x^2}{\frac{144}{25}} - \frac{y^2}{\frac{81}{25}} = 1$

$$\text{Foci} = \left(\pm \sqrt{16 - b^2}, 0 \right), \left(\pm \sqrt{\frac{144}{25} + \frac{81}{25}}, 0 \right)$$

$$\text{Same } \therefore 16 - b^2 = \frac{225}{25}$$

$$\Rightarrow 16 - b^2 = 9 \Rightarrow b^2 = 7$$

Maths - Question 59

Original paper page 15; original question and option regions

59. If (2, -1, 2) and (K, 3, 5) are the triads of direction ratios of two lines and the angle between them is 45° , then a value of K is

1) 2

2) 3

3) 4

4) 6

Sol:
$$\cos \theta = \frac{a_1 a_2 + b_1 b_2 + c_1 c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2} \sqrt{a_2^2 + b_2^2 + c_2^2}}$$

Maths - Question 59 (continued)

Original paper page 16; original question and option regions

$$\frac{1}{\sqrt{2}} = \frac{2k-3+10}{\sqrt{9}\sqrt{k^2+34}}$$

$$\Rightarrow 9(k^2+34) = 2(2k+7)^2$$

$$\Rightarrow k^2 - 56k + 208 = 0$$

$$\Rightarrow k = 4 \text{ or } 52$$

Maths - Question 60

Original paper page 16; original question and option regions

60. The length of perpendicular from the origin to the plane which makes intercepts $\frac{1}{3}, \frac{1}{4}, \frac{1}{5}$ respectively on the coordinate axes is

1) $\frac{1}{5\sqrt{2}}$

2) $\frac{1}{10}$

3) $5\sqrt{2}$

4) 5

Sol: Equation of plane is $\frac{x}{\frac{1}{3}} + \frac{y}{\frac{1}{4}} + \frac{z}{\frac{1}{5}} = 1 \Rightarrow 3x + 4y + 5z = 1$

Distance from origin to the plane = $\frac{|1|}{\sqrt{9+16+25}} = \frac{1}{\sqrt{50}} = \frac{1}{5\sqrt{2}}$

Maths - Question 61

Original paper page 16; original question and option regions

61. Match the following :

- I) The centroid of the triangle formed by
(2, 3, -1), (5, 6, 3), (2, -3, 1) is

a) (2, 2, 2)
- II) The circumcentre of the triangle formed by
(1, 2, 3), (2, 3, 1), (3, 1, 2) is

b) (3, 1, 4)
- III) The orthocentre of the triangle formed by
(2, 1, 5), (3, 2, 3), (4, 0, 4) is

c) (1, 1, 0)
- IV) The incentre of the triangle formed by
(0, 0, 0), (3, 0, 0), (0, 4, 0) is

d) (3, 2, 1)
- e) (0, 0, 0)

	I	II	III	IV		I	II	III	IV
1)	d	a	b	c	2)	a	b	c	d
3)	d	e	b	c	4)	d	a	e	c

Sol: I.G = (321)

II. Equilateral triangle S = G = (2,2,2)

III. Equilateral triangle H = G = (3,1,4)

IV. I = (1,1,0)

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Maths - Question 61 (continued)

Original paper page 17; original question and option regions

$$\frac{1}{x^2} = \frac{1}{x^2} \quad \frac{1}{x^2} = \frac{1}{x^2}$$

Maths - Question 62

Original paper page 17; original question and option regions

62. If $g(x) = \frac{x}{[x]}$ for $x > 2$ then $\lim_{x \rightarrow 2^+} \frac{g(x) - g(2)}{x - 2} =$

1) -1

2) 0

3) $\frac{1}{2}$

4) 1

Sol: $\lim_{x \rightarrow 2} \frac{\frac{x}{x} - 1}{x - 2} = \frac{1}{2}$

(ans -)

Maths - Question 63

Original paper page 17; original question and option regions

63. $\lim_{x \rightarrow \frac{\pi}{2}} \left(\frac{\angle X - \pi}{\cos X} \right) =$

1) 0

2) 1/2

3) -2

4) 5

Sol: apply L¹ H rule

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{2}{-\sin x} = -2$$

$$f(x) = \begin{cases} x & \text{for } 0 < x < 1 \end{cases}$$

Maths - Question 64

Original paper page 17; original question and option regions

64. If f is defined by $f(x) = \begin{cases} 2-x, & \text{for } x \geq 1 \\ 2-x^2, & \text{for } x < 1 \end{cases}$, then at $x = 1$, f is

- | | |
|-------------------------------------|--|
| 1) Continuous and differentiable | 2) Continuous but not differentiable |
| 3) Discontinuous but differentiable | 4) Neither continuous nor differentiable |

Sol: $\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} (2-x^2) = 2-1 = 1$ $\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} (2-x) = 2-1 = 1$

$$f(1) = 1$$

$$\therefore \lim_{x \rightarrow 1} f(x) = f(1)$$

$\therefore f$ is continuous at $x = 1$

$$f'(x) = \begin{cases} -2x & \text{if } 0 < x < 1 \\ -1 & \text{if } x > 1 \end{cases}$$

$$f'(1+) \neq f'(1-) \therefore f \text{ is not differentiable at } x = 1$$

Maths - Question 65

Original paper page 17; original question and option regions

65. If $x^2 + y^2 = t + \frac{1}{t}$ and $x^4 + y^4 = t^2 + \frac{1}{t^2}$ then $\frac{dy}{dx} =$

1) $\frac{-x}{y}$

2) $\frac{-y}{x}$

3) $\frac{x^2}{y^2}$

4) $\frac{y^2}{x^2}$

Sol: $x^2 + y^2 = t + \frac{1}{t}$

So BS

$$x^4 + y^4 + 2x^2y^2 = t^2 + \frac{1}{t^2} + 2$$

$$x^2y^2 = 1$$

Different wrt 'x'

$$\frac{dy}{dx} = -y/x$$

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Maths - Question 65 (continued)

Original paper page 18; original question and option regions

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Maths - Question 66

Original paper page 18; original question and option regions

- 66. Let D be the domain of a twice differentiable function f . For all $x \in D$, $f''(x) + f(x) = 0$ and $f(x) = \int g(x)dx + \text{constant}$. If $h(x) = (f(x))^2 + (g(x))^2$ and $h(0) = 5$ then $h(2015) - h(2014) =$**
- 1) 5 2) 3 3) 0 4) 1

Sol: $h(x)$ is constant function $\forall x \in \mathbb{R}$

Since $h(0) = 5$

$h(2015) = 5$

$h(2014) = 5$

now $h(2015) - h(2014) = 5 - 5 = 0$

Maths - Question 67

Original paper page 18; original question and option regions

67. If $x = at^2$ and $y = 2at$, then $\frac{d^2y}{dx^2}$ at $t = \frac{1}{2}$ is

1) $\frac{-2}{a}$

2) $\frac{4}{a}$

3) $\frac{8}{a}$

4) $\frac{-4}{a}$

Sol: $\frac{dx}{dt} = 2at, \frac{dy}{dt} = 2a$

$$\frac{dy}{dx} = \frac{1}{t} \Rightarrow \frac{d^2y}{dx^2} = \frac{-1}{t^2} \cdot \frac{1}{2at} \text{ at } t = \frac{1}{2}$$

$$= -\frac{4}{a}$$

Maths - Question 68

Original paper page 18; original question and option regions

68. The volume of a sphere is increasing at the rate of 1200 c.cm/sec. The rate of increase in its surface area when the radius is 10 cm is

1) 120 sq. cm/sec

2) 240 sq. cm/sec

3) 200 sq. cm/sec

4) 100 sq. cm/sec

Sol: $\frac{dv}{dt} = 1200, r = 10$

$$V = \frac{4}{3}\pi r^3$$

$$\Rightarrow \frac{dr}{dt} = \frac{3}{\pi}$$

$$S = 4\pi r^2$$

$$\frac{dS}{dt} = 4\pi \cdot 2r \frac{dr}{dt}$$

$$= 4\pi \cdot 2 \cdot 10 \cdot \frac{3}{\pi}$$

$$= 240 \text{ sq. cm/sec}$$

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Maths - Question 68 (continued)

Original paper page 19; original question and option regions

x 14

Maths - Question 69

Original paper page 19; original question and option regions

69. The slope of the tangent to the curve $y = \int_0^x \frac{dt}{1+t^3}$ at the point where $x = 1$ is

1) $\frac{1}{4}$

2) $\frac{1}{3}$

3) $\frac{1}{2}$

4) 1

Sol: Diff b.s

$$\frac{dy}{dx} = \frac{1}{1+x^3} = \frac{1}{2}$$

At $x = 1$

Maths - Question 70

Original paper page 19; original question and option regions

70. If $x^2 + y^2 = 25$, then $\log_5[\text{Max}(3x + 4y)]$ is

1) 2

2) 3

3) 4

4) 5

Sol: $x = 3, y = 4$

$$\log_5(\max(3x + 4x)) = \log_5^{25} = 2$$

Maths - Question 71

Original paper page 19; original question and option regions

71. If f is defined in $[1, 3]$ by $f(x) = x^3 + bx^2 + ax$, such that $f(1) - f(3) = 0$ and $f'(c) = 0$ where

$$c = 2 + \frac{1}{\sqrt{3}}, \text{ then } (a, b) =$$

- 1) $(-6, 11)$ 2) $\left(2 - \frac{1}{\sqrt{3}}, 2 + \frac{1}{\sqrt{3}}\right)$ 3) $(11, -6)$ 4) $(6, 11)$

Sol: since $f(1) = f(3)$

$$\Rightarrow 1 + b + a = 27 + 9b + 3a \Rightarrow 8b + 2a = -2b$$

From options $a = 11, b = -6$, satisfies

Maths - Question 72

Original paper page 19; original question and option regions

72. $\int \frac{ux}{(x-1)\sqrt{x^2-1}} =$

1) $-\sqrt{\frac{x-1}{x+1}} + c$

2) $\sqrt{\frac{x-1}{x^2+1}} + c$

3) $-\sqrt{\frac{x+1}{x-1}} + c$

4) $-\sqrt{\frac{x^2+1}{x-1}} + c$

(c is a constant)

Sol: Rmt $x-1 = \frac{1}{t} \Rightarrow dx = -\frac{1}{t^2} dt$

$$G.I = - \int (1+2t)^{\frac{-1}{2}} dt = -\sqrt{\frac{x+1}{x-1}} + c$$

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Maths - Question 72 (continued)

Original paper page 20; original question and option regions

$$x^2 + 1$$

Maths - Question 73

Original paper page 20; original question and option regions

73. $\int e^x \frac{x+1}{(x+1)^2} dx =$

1) $\frac{e^x}{x+1} + c$

2) $\frac{-e^x}{x-1} + c$

3) $e^x \left(\frac{x-1}{x+1} \right) + c$

4) $e^x \left(\frac{x+1}{x-1} \right) + c$

Sol: G.I $\int e^x \left[\frac{x^2 - 1 + 2}{(x+1)^2} \right] dx$

$$= \int e^x \left(\frac{x-1}{x+1} + \frac{2}{(x+1)^2} \right) dx = e^x / \left(\frac{x-1}{x+1} \right) + c$$

$$= \text{use } \int e^x [f(x) + f'(x)] dx = e^x \cdot f(x) + c$$

Maths - Question 74

Original paper page 20; original question and option regions

74. $\int \frac{x+1}{x(1+xe^x)} dx =$

1) $\log \left| \frac{1+xe^x}{xe^x} \right| + c$

2) $\log \left| \frac{xe^x}{1+xe^x} \right| + c$

3) $\log |xe^x(1+xe^x)| + c$

4) $\log (1+xe^x) + C$

Sol: Sub $1+xe^x = t$

$$G.I = \int \frac{1}{t(t-1)} dt$$

$$= \int \left(-\frac{1}{t} + \frac{1}{t-1} \right) dt$$

$$= \log \left| \frac{t-1}{t} \right|$$

$$= \log \left| \frac{xe^x}{1+xe^x} \right| + c$$

$$f(x) = f'(x) \quad f'(x) = f(x)$$

Maths - Question 75

Original paper page 20; original question and option regions

75. $\int \frac{f(x)g'(x) - f'(x)g(x)}{f(x)g(x)} [\log(g(x)) - \text{Log}(f(x))] dx =$

1) $\log\left(\frac{g(x)}{f(x)}\right) + C$

2) $\frac{1}{2} \left[\log\left(\frac{g(x)}{f(x)}\right) \right]^2 + C$

3) $\frac{g(x)}{f(x)} \text{Log}\left(\frac{g(x)}{f(x)}\right) + C$

4) $\text{Log}\left[\frac{g(x)}{f(x)}\right] - \frac{g(x)}{f(x)} + C$

(C is a constant)

Maths - Question 75 (continued)

Original paper page 21; original question and option regions

Sol: Sub $\log \left(\frac{g(x)}{f(x)} \right) = t$

$$G.I = \int t \, dt = \frac{t^2}{2} = \frac{1}{2} \left[\log \left(\frac{g(x)}{f(x)} \right) \right]^2 + c$$

$$\frac{\pi}{4} \sin x + \cos x$$

Maths - Question 76

Original paper page 21; original question and option regions

76. $\int_0^{\frac{\pi}{2}} \frac{\sin x + \cos x}{3 + \sin 2x} dx =$

1) $\frac{1}{2} \text{Log} 3$

2) $\text{Log} 2$

3) $\text{Log} 3$

4) $\frac{1}{4} \text{Log} 3$

Sol: Sub $\sin x - \cos x = t$

$$G.I = \int_{-1}^0 \frac{1}{2^2 - t^2} dt = \frac{1}{4} \log 3$$

$$1 \quad \sqrt{1 - \sin^2 x} \quad \sqrt{1 - \cos^2 x}$$

Maths - Question 77

Original paper page 21; original question and option regions

77. $\int_{-1}^1 \frac{\sqrt{1+x+x^2} - \sqrt{1-x+x^2}}{\sqrt{1+x+x^2} + \sqrt{1-x+x^2}} dx =$

1) $\frac{3\pi}{2}$

2) $\frac{\pi}{2}$

3) 0

4) -1

Sol: Odd function. Use $\int_{-a}^a f(x) dx = 0$ if $f(x)$ is odd function

G.I = 0

Maths - Question 78

Original paper page 21; original question and option regions

78. The area of the region described by $\{(x,y)/x^2+y^2 \leq 1 \text{ and } y^2 \leq 1-x\}$ is

1) $\frac{\pi}{2} - \frac{2}{3}$

2) $\frac{\pi}{2} + \frac{2}{3}$

3) $\frac{\pi}{2} + \frac{4}{3}$

4) $\frac{\pi}{2} - \frac{4}{3}$

Sol: Area = circle area + $2 \int_0^1 (1-y^2) dy$

$$= \frac{\pi}{2} + 2 \left(y - \frac{y^3}{3} \right)_0^1$$

$$= \frac{\pi}{2} + 2 \left(\frac{2}{3} \right) = \frac{\pi}{2} + \frac{4}{3}$$

Maths - Question 79

Original paper page 21; original question and option regions

79. The solution of $\frac{dy}{dx} + \frac{1}{x} = \frac{e^y}{x^2}$ is

- 1) $2x = (1 + Cx^2)e^y$ 2) $x = (1 + Cx^2)e^y$ 3) $2x^2 = (1 + Cx^2)e^{-y}$ 4) $x^2 = (1 + Cx^2)e^{-y}$

(C is a constant)

Maths - Question 79 (continued)

Original paper page 22; original question and option regions

Sol: B.D.E

$$\div e^y$$

$$e^{-y} \frac{dy}{dx} + \frac{1}{x} \cdot e^{-y} = \frac{1}{x^2}$$

Part $e^{-y} = t$

$$-e^{-y} \frac{dy}{dx} = \frac{dt}{dx}$$

$$\text{D.E is } \frac{dt}{dx} - \frac{1}{x}t = \frac{1}{x^2}$$

$$\text{I.F} = e^{\int -\frac{1}{x} dx} = e^{-\log x} = \frac{+1}{x}$$

$$\text{G.S. is } \frac{1}{e^y} \cdot \frac{1}{x} = \int \frac{-1}{x^2} \cdot \frac{1}{x} dx$$

$$\frac{1}{e^y x} = \frac{1}{2x^2} + c$$

$$= 2x = e^y (1 + cx^2)$$

Maths - Question 80

Original paper page 22; original question and option regions

80. The differential equation $\frac{dy}{dx} = \frac{1}{ax + by + c}$ where a, b, c are all non zero real numbers, is

1) Linear in y

2) Linear in x

3) Linear in both x & y

4) Homogeneous equation

Sol: $\frac{dx}{dy} = ax + by + c$

$$\frac{dx}{dy} - ax = by + c$$

L.D.E in (x)

AP EAMCET / EAPCET | 2015 | Shift not identified | Engineering

Maths - Question 80 (continued)

Original paper page 23; original question and option regions

PHYSICS